SPS-24G-01

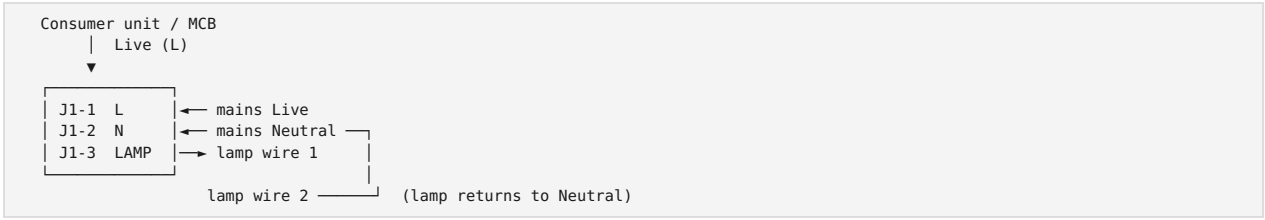
SPS-24G-01 — Application & Installation Note

SAMAY ELECTRONICS · 24 GHz Smart Presence Sensor Switch · Rev A0

> ⚠ Installation involves **230 V mains**. A licensed electrician must do it, with the circuit breaker OFF. This product is not yet safety-certified — see `SAFETY_REVIEW.md`.

1. Wiring

The switch needs **Line, Neutral and the Lamp wire** (a neutral-required smart switch).



- **J1-1 (L):** incoming live from the MCB.
- **J1-2 (N):** neutral — powers the electronics even when the lamp is off.
- **J1-3 (LAMP):** switched live to the light. The light's other terminal connects to Neutral.

Use correctly rated conductors and ferrules; torque the screw terminals to spec; sleeve/insulate before closing the modular plate. Keep the low-voltage (bottom) half of the board away from any exposed metal.

2. Mounting & orientation

- Fits a standard 86 × 86 mm modular box. Secure the PCB to the modular frame via the four M3 holes.
- The **LD2410B radar stands vertical on header J3**, antenna facing the **front cover** (the room).
- Mount height: **wall 1.1–1.5 m** or **ceiling 2.4–3 m**. Avoid aiming through thin walls into corridors.
- The front cover must be **RF-transparent plastic** with **no metallised paint or foil** over the antenna.

3. Commissioning

1. With mains OFF, fit the module and close the plate. Restore power.
2. The LED shows the state (solid = lamp on / presence, slow blink = manual, fast blink = fault).
3. **Off-delay:** adjust trimmer **RV1** — CCW ≈ 10 s, CW ≈ 10 min.
4. **Range / sensitivity:** plug a 3.3 V USB-UART into **J4** (TX/RX/GND) and run the Hi-Link LD2410B PC tool. Set the furthest detection gate (0.75 m per gate, gate 8 = 6 m) and per-gate sensitivity so the switch ignores motion outside the room. Settings persist in the radar.
5. **Manual override (SW1):** short press = Auto↔Manual (and toggle in Manual); long press = force OFF.

4. Load compatibility

Load	Notes
LED lamp / driver	On-board snubber fitted. If a low-power (<3 W) driver flickers when "off", add a 0.1 µF X2 + 100 kΩ bleeder across the lamp, or use the relay-output variant.
Incandescent / halogen	OK to ~2 A without heatsink; fit a TO-220 clip heatsink for 2–6 A.
CFL	Works, but CFLs dislike rapid cycling — set a longer off-delay.
Inductive (fan)	Keep the snubber; prefer the snubberless BTA16-600BW for heavy inductive loads.

Minimum load ~3 W for reliable triac latching.

5. Troubleshooting

Symptom	Likely cause / action
Light never turns on	Check Lamp wiring (J1-3), minimum load, radar OUT at TP4, triac gate TP3.
Light won't turn off	Radar seeing motion outside the room → reduce range/sensitivity via J4; check off-delay (RV1).
Flicker when off	Low-power LED driver leakage → add bleeder / dummy load across the lamp.
Fast-blink LED	Radar not responding → check 5 V at the header, radar seated, UART config.
No LED, dead	Check F1 fuse, HLK-PM01 5 V at TP1. **De-energise first.**

6. Firmware / programming

Program the ATtiny13A via the **J2** 6-pin AVR ISP header with an ISP programmer (USBasp etc.), **mains disconnected**. Reference firmware and pin map are in `../06_firmware/`.